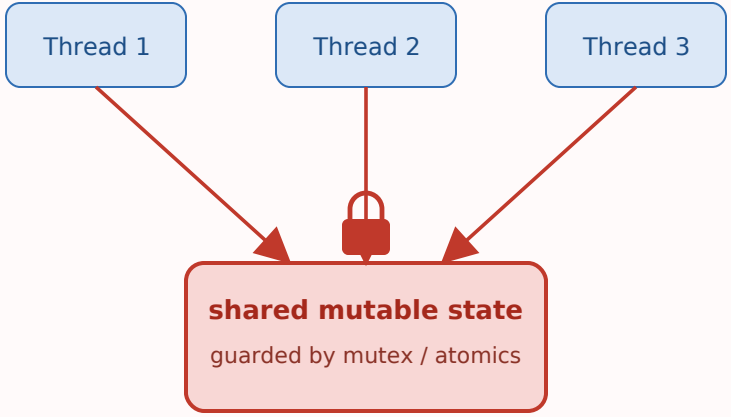
Two concurrency models for control-plane state

Shared-memory concurrency

threads mutate common state behind locks



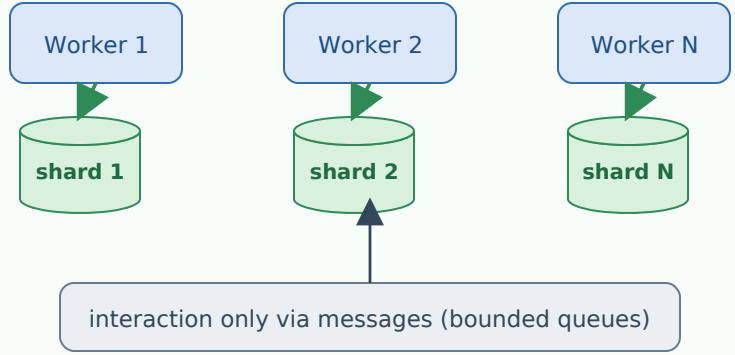
Hazards:

data races · deadlock · lock contention
priority inversion · hard-to-reason worst case

worst-case timing is hard to bound

Shared-nothing (message-passing)

state partitioned, each piece owned by one thread



Benefits:

no shared mutable state · no locks on data
simpler reasoning · scales · bounded worst case

chosen for BOTH implementations (maps to per-UE state)